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Centre Number

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Student Number



SCEGGS DARLINGHURST

TRIAL EXAMINATION 2001

Mathematics

Year 12

2/3 Unit

TIME ALLOWED: 3 hours (plus 5 minutes reading time)

This is a TRIAL PAPER only and does not necessarily reflect the content or format of the Higher School Certificate Examination for this subject.

ASSESSMENT WEIGHTING 40%

Directions to Candidates:

- Attempt all ten questions. All questions are of equal value.
- Ensure that your student number is on this paper.
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Approved scientific calculators should be used. Mathematical templates and geometrical instruments may be used.
- Begin each question on a new page - write your number at the top of each page.
- The table of standard integrals is printed on the last page.

Question 1. (12 Marks)

Marks

(a) Factorise:

(i) $4y^2 - 100$

1

(ii) $k^3 - 8$

1

(b) Rationalise $\frac{8}{3 - \sqrt{5}}$

2

(c) Solve $\frac{2x}{3} - \frac{x+1}{4} = 1$

2

(d) Express $0.0\dot{6}$ as a fraction in simplest form

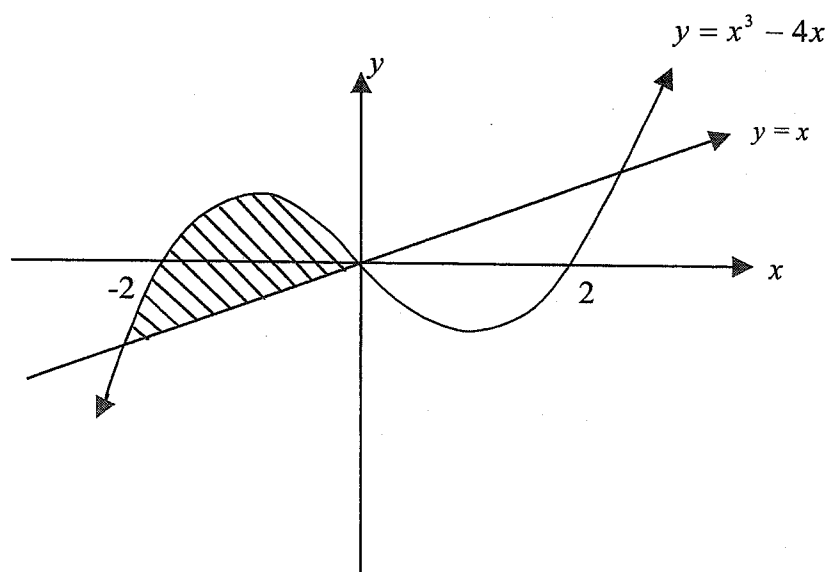
2

(e) Solve $x^2 + 4x + 3 \geq 0$

2

(f) Write a pair of inequalities to describe the shaded region:

2



Question 2. (12 Marks)

Start a new page.

Marks

Consider $A(-6, -5)$ and $B(8, 9)$

- | | |
|--|---|
| (a) Calculate the distance AB, in simplest form | 1 |
| (b) Find the equation of the line joining A and B in general form | 3 |
| (c) Find C, the midpoint of AB | 1 |
| (d) Calculate the perpendicular distance of $D(0, 3)$ from the line AB | 1 |
| (e) Show that DC has the equation $x + y - 3 = 0$ | 2 |
| (f) Write down the co-ordinates of E such that E lies on DC produced and $DC = CE$ | 1 |
| (g) What sort of quadrilateral is AEBD?
Justify your answer | 2 |
| (h) Hence, find the area of AEBD | 1 |

Question 3. (12 Marks)

Start a new page

Marks

- (a) 20 identical cards numbered 1 to 20 are placed in a bag, and one is drawn out at random.

2

What is the probability that this card will be less than 10 or divisible by 4?

- (b) Differentiate:

(i) $y = x^2 e^{3x}$

2

(ii) $y = \frac{\sin 3x}{x}$

2

- (c) Find

(i) $\int \frac{dx}{x+5}$

2

(ii) $\int \sec^2 5x \, dx$

2

- (d) Prove that $\frac{\sin \theta}{\cos \theta} (1 - \cot^2 \theta) + \frac{\cos \theta}{\sin \theta} (1 - \tan^2 \theta) = 0$

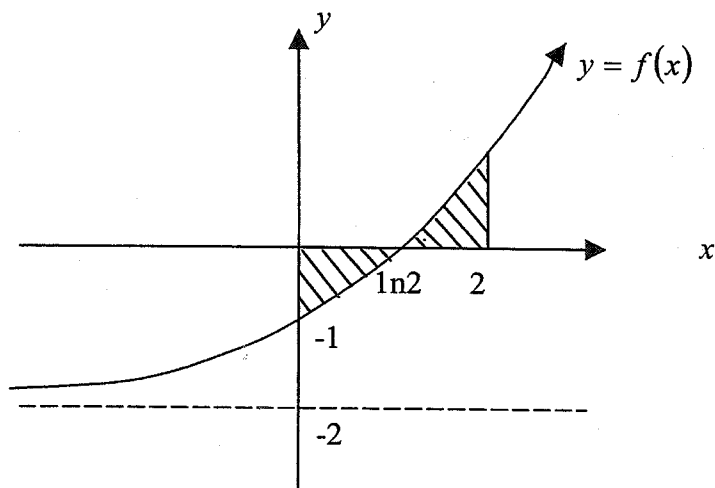
2

Question 4 (12 Marks)

Start a new page

Marks

(a)



(i) Write an expression to describe the shaded area.

2

(ii) Give a possible equation for $y = f(x)$

1

(b) Evaluate $\sum_{k=2}^6 3^k$

2

(c) Show that the equation of the tangent to the curve $y = \ln(x^2 + 3)$ at the point $x = 3$ is $x - 2y + \ln 144 - 3 = 0$

2

(d) A function $y = f(x)$ has a stationary point at $(1, 2)$ and $f''(x) = 6x - 2$

Find:

(i) the equation of the function.

3

(ii) the co-ordinates of the other stationary point and determine its nature.

2

Question 5. (12 Marks)

Start a new page

Marks

- (a) The table shows the values of a function $f(x)$ for five x values.

3

x	1	1.5	2	2.5	3
$f(x)$	1.011	1.179	1.322	1.447	1.559

Approximate the value of $\int_1^3 f(x) dx$ using the five values and Simpson's Rule.

- (b) Find the values of m for which the equation $x^2 + 4mx + 8 - 4m = 0$ has equal roots

2

- (c) Consider the function $f(x) = x^3 - 12x + 16$

- (i) Find the co-ordinates of any stationary points and determine their nature.

2

- (ii) Find any points of inflexion

2

- (iii) Sketch the curve $y = f(x)$, $-5 \leq x \leq 3$

2

- (iv) Find the minimum value of the curve in this domain.

1

Question 6. (12 Marks)

Start a new page

Marks

(a) Consider the curve $(x - 4)^2 = 16(y - 2)$

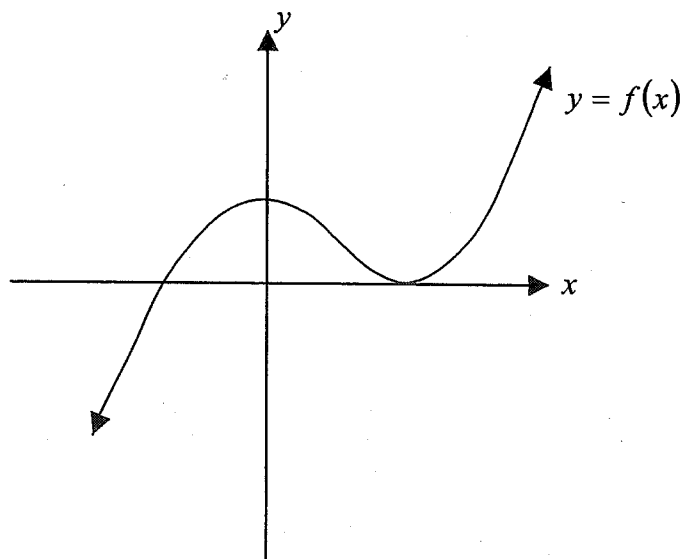
(i) Write down the co-ordinates of the focus.

2

(ii) State the equation of the directrix.

1

(b)



Copy this diagram onto your answer page.

(i) On the same set of axes, sketch $y = f'(x)$

1

(ii) On the same set of axes, using a dotted line, sketch a primitive of $y = f(x)$

2

(c) The sum of the height, h cm, of a cylinder and the circumference of its base of radius r cm, is 10cm.

(i) Show that $h = 10 - 2\pi r$

1

(ii) Show that the volume of the cylinder is $V = \pi r^2 (10 - 2\pi r)$

1

(iii) Find the exact value of r at which the volume of the cylinder is a maximum

3

(iv) Hence find the maximum volume of the cylinder, correct to 2 decimal places.

1

Question 7. (12 marks)

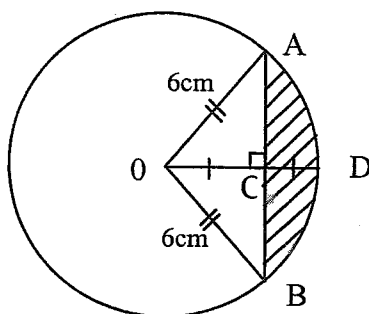
Start a new page

Marks

(a) Solve $e^{2x} - 4e^x - 5 = 0$.

3

(b)



O is the centre of the circle.
Find the exact value of the shaded area.

3

- (c) (i) Sketch the curve $y = -\sin(2x)$ and $y = \frac{x}{3}$ on the same axes in the domain $0 \leq x \leq \pi$

2

- (ii) Hence, state the number of solutions to the equation $-\sin(2x) = \frac{x}{3}$ in the domain $0 \leq x \leq \pi$

1

- (iii) One solution was approximated as 1.9.
Use this approximation to calculate the area between the two functions in the domain $0 \leq x \leq 1.9$ correct to 1 decimal place.

3

Question 8. (12 Marks)

Start a new page

Marks

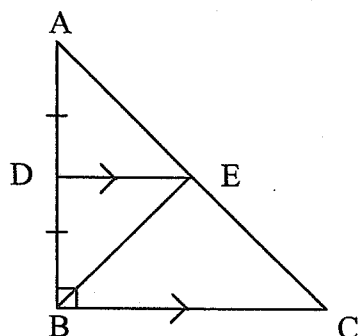
- (a) Sally and Brett decide to buy a new house for \$300000. They agree to pay back the loan in equal monthly instalments of \$ P over 20 years.

Interest is charged at 6% p.a., compounded monthly.

- (i) Show that the amount Sally and Brett owe at the end of the second month is: $A_2 = 300000(1.005)^2 - P(1 + 1.005)$ 2

- (ii) Hence, calculate the value of each monthly instalment, \$ P . 3

- (b) Triangle ABC has a right angle at B, D is the midpoint of AB, and $DE \parallel BC$.



- (i) Find the size of $\angle ADE$, giving reasons 1

- (ii) Prove that $\triangle AED \equiv \triangle BED$ 2

- (iii) Prove that $BE = EC$ 2

- (c) Sketch a curve that has all of the following properties: 2

$$f(2) = 1$$

$$f'(2) = 0$$

$$f''(2) = 0$$

$$f'(x) \geq 0 \text{ for all real } x$$

Question 9. (12 Marks)

Start a new page

Marks

- (a) The percentage of the Mathematics course that is understood by the students in October is given by the equation:

$$P = \frac{-x^2}{1280} (4x - 240)$$

where x is the number of revision assignments completed and P is the percentage of the course that is understood.

- | | | |
|---------|--|---|
| (i) | Show that a student completing 40 assignments understands 100% of the course. | 2 |
| (ii) | With reference to a graph, or otherwise, explain what happens when $x > 40$ | 2 |
| (iii) | Find the value of x , $0 \leq x \leq 40$ for which the percentage of the course being understood increases at the greatest rate. | 2 |
| (b) (i) | On the same set of axes, sketch $y = e^x + 1$ and $y = e + 1$ | 1 |
| (ii) | Find the co-ordinates of the single point of intersection. | 1 |
| (iii) | The area between $y = e + 1$, the y -axis and $y = e^x + 1$ is rotated about the x -axis. | 4 |

Find the volume of the solid of revolution formed.

Question 10. (12 Marks)

Start a new page.

Marks

- (a) A particle is moving in a straight line.
At time, t seconds, its distance, x metres, from a fixed point 0 is given by

$$x = 2 + 2 \sin \frac{\pi}{4} t$$

- (i) Calculate its initial position. 1
- (ii) Find its exact velocity after 1 second. 2
- (iii) Find the position of the particle when the particle is stationary for the second time. 2
- (b) (i) Draw a table showing the outcome of rolling two dice and recording their sum. 1
- (ii) Hence, copy and complete the following table of probabilities. 4

P (even number closer to 5 rather than 10)	
P (even number closer to 10 rather than 5)	
P (odd number closer to 5 rather than 10)	
P (odd number closer to 10 rather than 5)	

- (iii) Year 2 students at SCEGGS play a game called Banko. 2
- Two players, A and B, each begin with 18 blue counters and 12 red counters. The teacher rolls two dice and the sum is noted.
- If the sum of the numbers is even, then Player A removes one of Player B's counters.
- If the sum of the numbers is odd, then Player B removes one of Player A's counters.
- The colour of the counter to be removed is determined as follows:
- a red counter is removed if the sum is closer to 5 than 10.
 - a blue counter is removed if the sum is closer to 10 than 5.
- The winner is the first person to remove all the other player's counters.
- Use your results from part (ii) to explain whether or not this game is fair.

END OF PAPER

MATHEMATICS TRIAL
SOLUTIONS, 2001

QUESTION 1: (12 marks)

(a) (i) $4(y^2 - 25)$

$$= 4(y+5)(y-5)$$

(ii) $(k-2)(k^2 + 2k + 4)$

(b)
$$\frac{8}{3-\sqrt{5}} \times \frac{3+\sqrt{5}}{3+\sqrt{5}} = \frac{24+8\sqrt{5}}{9-5}$$
$$= 6+2\sqrt{5}$$

(c)
$$\frac{2x}{3} - \frac{x+1}{4} = 1$$

$$8x - 3(x+1) = 12$$

$$5x - 3 = 12$$

$$5x = 15$$

$$\therefore x = 3$$

(d) Let $x = 0.06666\dots$

$$\therefore 10x = 0.6666\dots$$

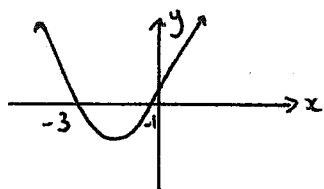
$$100x = 6.666\dots$$

$$\therefore 90x = 6$$

$$x = \frac{6}{90} = \frac{1}{15}$$

(e) $x^2 + 4x + 3 \geq 0$

$$(x+3)(x+1) \geq 0$$



$$\therefore x \leq -3 \text{ and } x \geq -1$$

(f) $y \leq x^3 - 4x$

and $y \geq x$

Com
2

QUESTION 2: (12 marks)

(a) $AB = \sqrt{(8--6)^2 + (9--5)^2}$

$$= \sqrt{196+196}$$

$$= \sqrt{392}$$

$$= 14\sqrt{2}$$

(b) $m_{AB} = \frac{9--5}{8--6} = 1$

$$y - 9 = 1(x - 8)$$

$$y = x + 1$$

$$\therefore x - y + 1 = 0$$

(c) $C(1, 2)$

(d)
$$d = \frac{|0 - 3 + 1|}{\sqrt{1^2 + 1^2}}$$

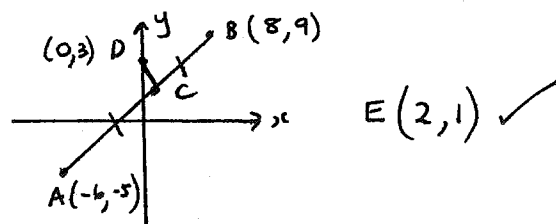
$$= \frac{2}{\sqrt{2}} \quad \checkmark \quad (= \sqrt{2})$$

(e) $m_{DC} = \frac{3-2}{0-1} = -1$

$$\therefore y - 3 = -1(x - 0)$$

$$x + y - 3 = 0$$

(f)

(g) Diagonals bisect each other at right angles $\therefore$ Rhombus. Leas
2

(h) $\frac{1}{2} \times 14\sqrt{2} \times 2\sqrt{2} = 28 \text{ units}^2$

QUESTION 3: (12 marks)

(a) $\frac{\sqrt{9+3}}{20} = \frac{3}{5} \checkmark$

(b) (i) $u = x^2 \quad v = e^{3x}$
 $u' = 2x \quad v' = 3e^{3x}$

Calc 4 $\frac{dy}{dx} = 2xe^{3x} + 3x^2e^{3x}$
 $= xe^{3x}(2 + 3x) \checkmark$

(ii) $u = \sin 3x \quad v = x$
 $u' = 3\cos 3x \quad v' = 1 \checkmark$

$\frac{dy}{dx} = \frac{3x \cos 3x - \sin 3x}{x^2} \checkmark$

(c) (i) $\int \frac{dx}{x+5}$

Calc 4 $= \ln(x+5) + C \checkmark \checkmark +C$

(ii) $\int \sec^2 5x \, dx$
 $= \frac{1}{5} \tan 5x + C \checkmark \checkmark$

(d) LHS $= \frac{\sin \theta}{\cos \theta} (1 - \cot^2 \theta) + \frac{\cos \theta}{\sin \theta} (1 - \tan^2 \theta)$

$= \frac{\sin \theta}{\cos \theta} - \frac{\cos \theta}{\sin \theta} + \frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \checkmark$

Leas
2 $= 0$
 $= \text{RHS} \checkmark$

QUESTION 4: (12 marks)

(a) (i) Area $= \left| \int_0^{\ln 2} f(x) \, dx \right| + \int_{\ln 2}^2 f(x) \, dx$

(ii) $y = e^x - 2 \checkmark$ Com 3

(b) $\sum_{k=2}^6 3^k = 9 + 27 + 81 + 243 + 729$
 $= 1089$

or $\frac{9(1-3)}{1-3} = 1089 \checkmark \checkmark$

(c) $y = \ln(x^2 + 3)$

$\frac{dy}{dx} = \frac{2x}{x^2 + 3}$

At $x=3$, $\frac{dy}{dx} = \frac{6}{12} = \frac{1}{2} \checkmark$

At $x=3$, $y = \ln 12$

$\therefore y - \ln 12 = \frac{1}{2}(x - 3)$

$2y - 2\ln 12 = x - 3 \checkmark$

$\therefore x - 2y + 2\ln 12 - 3 = 0$

$x - 2y + \ln 144 - 3 = 0$

(d) (i) $f''(x) = 6x - 2$

$f'(x) = 3x^2 - 2x + C \checkmark$

But (1,2) is stat point

$\therefore 0 = 3 - 2 + C$

$\therefore C = -1$

$\therefore f'(x) = 3x^2 - 2x - 1 \checkmark$

$\therefore f(x) = x^3 - x^2 - x + C$

(1,2) lies on curve

$\therefore 2 = 1 - 1 - 1 + C$

$\therefore C = 3$

$\therefore f(x) = x^3 - x^2 - x + 3 \checkmark$

(ii) Stat point $\Rightarrow f'(x) = 0$

$$3x^2 - 2x - 1 = 0$$

$$(3x + 1)(x - 1) = 0$$

$$\therefore x = 1 \text{ or } -\frac{1}{3}$$

$$y = 2 \approx \frac{86}{27}$$

when $x = -\frac{1}{3}$, $f''(x) = -4$ ✓

$\therefore (-\frac{1}{3}, \frac{86}{27})$ is a max t.p.

QUESTION 5: (12 marks)

(a)

1	1.5	2	2.5	3
1.011	1.179	1.322	1.447	1.559
1	4	2	4	1
1.011	4.716	2.644	5.788	1.559
$\Sigma = 15.718$				✓

$$\text{Area} \doteq \frac{1}{3} \times \frac{1}{2} \times 15.718 \quad \checkmark$$

$$\doteq 2.62 \text{ units}^2 \text{ (2 dp)} \quad \checkmark$$

(b) Equal roots $\Rightarrow \Delta = 0$

$$\therefore (4m)^2 - 4 \times 1 \times (8 - 4m) = 0 \quad \checkmark$$

$$16m^2 + 16m - 32 = 0$$

$$m^2 + m - 2 = 0$$

$$(m + 2)(m - 1) = 0$$

$$\therefore m = -2 \text{ or } 1 \quad \checkmark$$

(c) (i) $f'(x) = 3x^2 - 12$

Stat. points occur $f'(x) = 0$ Calc 7

$$0 = 3x^2 - 12$$

$$\therefore x = \pm 2$$

$$y = 0, 32 \quad \checkmark$$

$$f''(x) = 6x$$

when $x = 2$, $f''(x) = 12 > 0$

$\therefore (2, 0)$ is min t.p.

when $x = -2$, $f''(x) = -12 < 0 \quad \checkmark$

$\therefore (-2, 32)$ is max t.p.

(ii) Possible p.o.i occur $f''(x) = 0$

$$6x = 0$$

$$\therefore x = 0$$

$$y = 16 \quad \checkmark$$

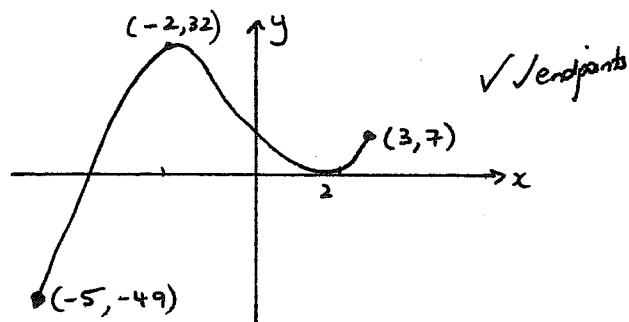
Test:

x	0^-	0	0^+
$f''(x)$	-ve	0	+ve

$\therefore$ Change in concavity ✓

$\therefore (0, 16)$ is a point of inflexion.

(iii)



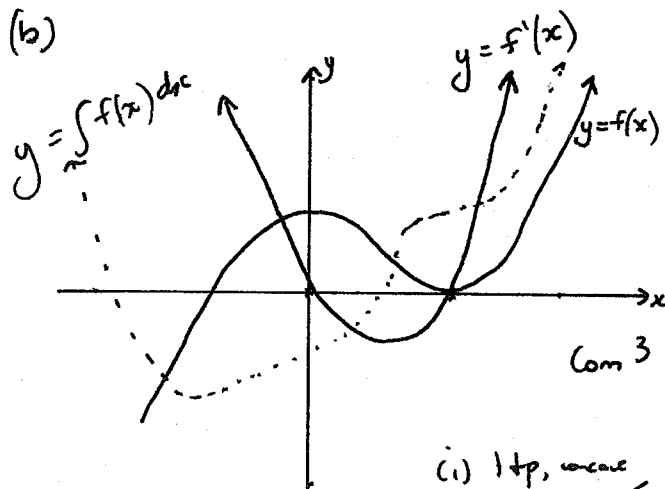
(iv) Min value is -49. ✓

QUESTION 6: (12 marks)

(a) (i) Vertex: $(4, 2)$ ✓
Focal length = 4

∴ Focus $(4, 6)$ ✓

(ii) $y = -2$ ✓



(i) 1 tp, increase up ✓

(ii) horizontal 10 and min ✓

(c) (i) $h + 2\pi r = 10$ ✓
∴ $h = 10 - 2\pi r$

(ii) $V = \pi r^2 h$ ✓
 $= \pi r^2 (10 - 2\pi r)$

(iii) $V = 10\pi r^2 - 2\pi^2 r^3$
 $\frac{dV}{dr} = 20\pi r - 6\pi^2 r^2$ ✓

$20\pi r - 6\pi^2 r^2 = 0$

$2\pi r (10 - 3\pi r) = 0$

∴ $r = 0$ and $r = \frac{10}{3\pi}$ ✓

$\frac{d^2V}{dr^2} = 20\pi - 12\pi^2 r$

When $r = \frac{10}{3\pi}$, $\frac{d^2V}{dr^2} < 0$

∴ $r = \frac{10}{3\pi}$ maximises volume. ✓

(iv) $V = 10\pi \left(\frac{10}{3\pi}\right)^2 - 2\pi^2 \left(\frac{10}{3\pi}\right)^3$

≈ 11.79 (2 dp) ✓

QUESTION 7: (12 marks)

(a) $e^{2x} - 4e^x - 5 = 0$

Let $u = e^x$

∴ $u^2 - 4u - 5 = 0$ ✓

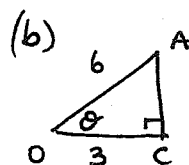
$(u-5)(u+1) = 0$

∴ $u = 5$ or -1 ✓

∴ $e^x = 5$ or $e^x = -1$
no solutions

∴ $x = \ln 5$ ✓

∴ $x = \ln 5$.



$\theta = \frac{1}{2}$

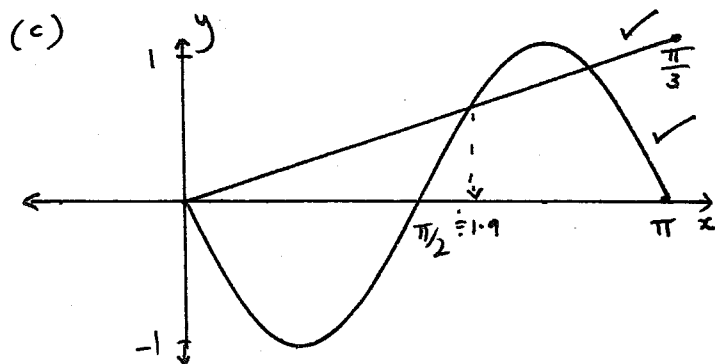
Keas
3

∴ $\theta = \frac{\pi}{3}$ ✓

∴ Area = $\frac{1}{2} \times 6^2 \times \left(\frac{2\pi}{3} - \sin \frac{2\pi}{3}\right)$ ✓

$= 12\pi - 18 \times \frac{\sqrt{3}}{2}$

$= 12\pi - 9\sqrt{3}$ ✓



(ii) 3 ✓

(iii) $A \doteq \int_0^{1.9} \frac{x}{3} + \sin 2x \, dx$ ✓

$$= \left[\frac{x^2}{6} - \frac{1}{2} \cos 2x \right]_0^{1.9}$$

$$= (0.602 + 0.395) - (0 - \frac{1}{2})$$

$$\doteq 1.497 \quad \checkmark$$

QUESTION 8: (12 marks)

(a) (i) $A_1 = 300,000 (1.005) - P$ ✓
 $A_2 = A_1 (1.005) - P$
 $= 300,000 (1.005)^2 - P(1 + 1.005)$ ✓

(ii) $A_3 = A_2 (1.005) - P$
 $= 300,000 (1.005)^3 - P(1 + 1.005 + 1.005^2)$

$\vdots$
 $A_n = 300,000 (1.005)^n - P(1 + 1.005 + \dots + 1.005^{n-1})$

But $A_{240} = 0$

$\therefore 0 = 300,000 (1.005)^{240} - P(1 + 1.005 + \dots + 1.005^{239})$

$$S_{240} = \frac{1(1.005^{240} - 1)}{0.005}$$

$$\doteq \$462.04 \quad \checkmark$$

$\therefore P = 300,000 (1.005)^{240} \div$

$\doteq \$2149.29 \quad \checkmark$

$\therefore$ Sally + Brett pay back each month.

(b) (i) $\angle ADE = 90^\circ$ (corresponding $\angle =$ as $DE \parallel BC$) ✓

(ii) DE is common

$\angle ADE = \angle BDE = 90^\circ$

$AD = DB$ (given) ✓✓

$\therefore \triangle AED \equiv \triangle BED$ (SAS)

(iii) $\angle AED = \angle BED$ (corresponding $\angle$ in $\triangle AED \equiv \triangle BED$) ✓

But $\angle AED = \angle ECB$ (corresponding $\angle =$ as $DE \parallel BC$)

leas
2

and $\angle BED = \angle EBC$ (alt. $\angle =$ as $DE \parallel BC$)

$\therefore \angle ECB = \angle EBC$

$\therefore BE = EC$ (sides opp $\angle$ in isosceles $\triangle BCE$) ✓

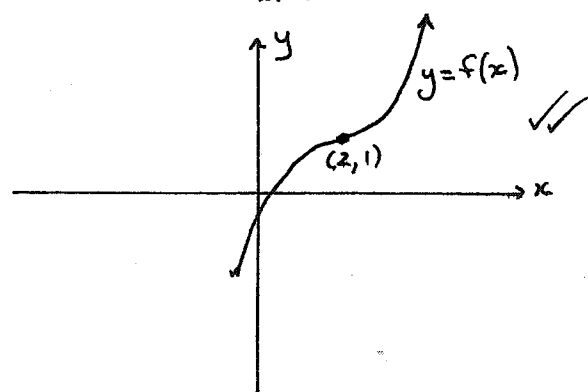
(c) $f(2) = 1 \Rightarrow$ passes through $(2, 1)$

$f'(2) = 0 \Rightarrow$ stat. point at $(2, 1)$

$f''(2) = 0 \Rightarrow$ neither concave up or down at $(2, 1)$ is point of inflexion

$f'(x) \geq 0 \Rightarrow$ curve increasing or stationary for all values of x

com
2



QUESTION 9: (12 marks)

(a) (i) $P = \frac{-x^2}{1280} (4x - 240)$

when $x = 40$,

$$P = \frac{-40^2}{1280} (160 - 240) \checkmark$$

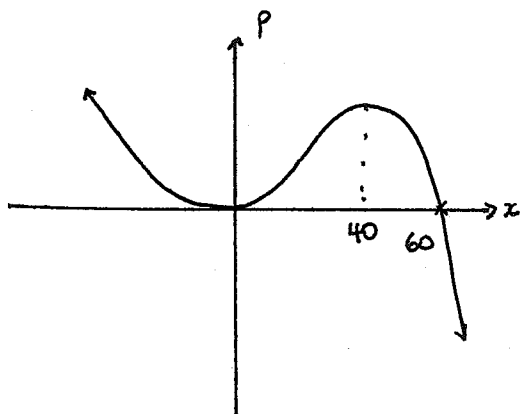
Calc

b

$$= 100 \checkmark$$

∴ A student completing 40 assignments understands 100% of the course.

(ii)



$$P = \frac{-4x^3}{1280} + \frac{240x^2}{1280}$$

$$= \frac{-x^3}{320} + \frac{3x^2}{16}$$

$$\frac{dP}{dx} = \frac{-3x^2}{320} + \frac{6x}{16} \checkmark$$

Stationary pts $\Rightarrow \frac{dP}{dx} = 0$

$$\text{i.e. } \frac{-3x^2}{320} + \frac{6x}{16} = 0$$

$$\frac{x}{320} (-3x + 120) = 0$$

$$\text{i.e. } x = 0 \text{ or } 40$$

∴ when $x > 40$, the percentage of the course understood decreases. $\checkmark$

(iii) $\frac{d^2P}{dx^2} = \frac{-6x}{320} + \frac{3}{8}$

$$= \frac{-3x}{160} + \frac{3}{8} \checkmark$$

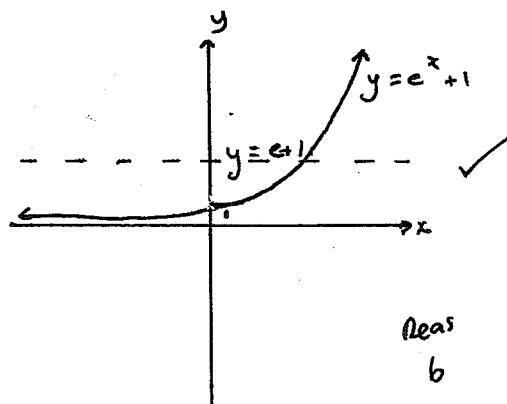
$$\frac{-3x}{160} + \frac{3}{8} = 0$$

$$\frac{3x}{160} = \frac{3}{8}$$

$$x = 20 \checkmark$$

∴ Increasing at the greatest rate when $x = 20$.

(b) (i)



(ii) $e + 1 = e^x + 1$

$$e = e^x$$

$$\therefore x = 1 \checkmark$$

$$y = e + 1$$

(iii) $V = \pi \int_0^1 (e+1)^2 dx - \pi \int_0^1 (e^x+1)^2 dx \checkmark$

$$= \pi [(e+1)^2 x]_0^1 - \pi \int_0^1 e^{2x} + 2e^x + 1 dx$$

$$= \pi (e+1)^2 - \pi \left[\frac{1}{2} e^{2x} + 2e^x + x \right]_0^1 \checkmark$$

$$= \pi (e+1)^2 - \pi \left[\left(\frac{1}{2} e^2 + 2e + 1 \right) - \left(\frac{1}{2} + 2 + 0 \right) \right] \checkmark$$

$$= \pi \left(e^2 + 2e + 1 - \frac{1}{2} e^2 - 2e + \frac{3}{2} \right) \checkmark$$

$$= \pi \left(\frac{1}{2} e^2 + \frac{5}{2} \right) = \frac{\pi}{2} (e^2 + 5) \mu^3 \checkmark$$

QUESTION 10: (12 marks)

(a) (i) $x = 2 + 2\sin\frac{\pi}{4} \times 0$

$\therefore x = 2. \checkmark$

(ii) $v = \frac{\pi}{4} \times 2\cos\frac{\pi}{4} t$

$= \frac{\pi}{2} \cos\frac{\pi}{4} t$ $\checkmark$

when $t = 1$, $v = \frac{\pi}{2} \cos\frac{\pi}{4}$

$= \frac{\pi}{2} \times \frac{1}{\sqrt{2}}$

$= \frac{\pi}{2\sqrt{2}} \left(\text{or } \frac{\sqrt{2}\pi}{4} \right) \checkmark$

Calc
5

(iii) Stationary $\Rightarrow v = 0$

$0 = \frac{\pi}{2} \cos\frac{\pi}{4} t$

$\therefore 0 = \cos\frac{\pi}{4} t$ $\checkmark$

$\therefore \frac{\pi}{4} t = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$

$\therefore t = 2, 6, 10, \dots$

$\therefore$ Stationary for the second time
at $t = 6. \checkmark$

(b)

(i)

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

$\checkmark$

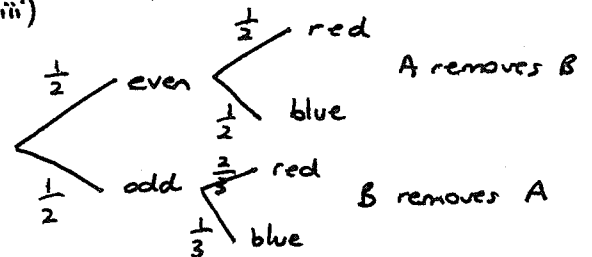
(ii) $P(\text{even, closer to 5}) = \frac{1}{2} \checkmark$

$P(\text{even, closer to 10}) = \frac{1}{2} \checkmark$ Reas

$P(\text{odd, closer to 5}) = \frac{2}{3} \checkmark$ 4

$P(\text{odd, closer to 10}) = \frac{1}{3} \checkmark$

(iii)



$\therefore$ Game is not fair. $\checkmark \checkmark$

Player A has a much higher
chance of removing 18 blue
counters ($\frac{1}{4}$ each time)
than Player B ($\frac{1}{6}$ each time) $\frac{6m}{2}$

PROFILES

Reasoning

Q 2 (g) - 2

Q 3 (d) - 2

Q 7 (b) - 3

Q 8 (b) (iii) - 2

Q 9 (b) - 6

Q 10 (b) (ii) - 4

19

Calc

Q 3 (b) + (c) - 8

Q 5 (c) - 7

Q 9 (a) - 6

Q 10 (a) - 5

26

Com

Q 1 (f) - 2

Q 4 (a) - 3

Q 6 (b) - 3

Q 8 (c) - 2

Q 10 (b) (ii) - 2

12